Hints: Methods of proof – 3. The extremal principle and compactness

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Existence: the convex body K is compact, so K^3 is compact, and the area of the triangle XYZ is a continuous function of $(X, Y, Z) \in K^3$. It is not identically zero since K has interior points.

Hint 2. For the maximal triangle ABC, fix A and B: the vertex C maximises the distance to the line AB over K. So K lies in the strip between the line AB and the line through C parallel to AB.

Hint 3. Thus the line through C parallel to AB is a supporting line of K at C (the tangent line if the boundary is smooth at C); the same holds for A and B. (Equivalently, K lies in the triangle with vertices $B + C - A$, $C + A - B$, $A + B - C$.)

Problem 2

Note. The statement needs at least five points: four points, three forming a triangle and one inside it, contain no convex quadrilateral. The hints prove the claim for five (hence for any $\geqslant 5$) points in general position.

Hint 1. Take five of the points and look at their convex hull. If it is a quadrilateral or a pentagon, you are done at once (for a pentagon take any four vertices).

Hint 2. Otherwise the hull is a triangle ABC with two points D, E inside. Draw the line DE; it misses all vertices (no three collinear), so two vertices of the triangle lie on the same side of it.

Hint 3. If A, B are on the same side of DE, then A, B, D, E is a convex quadrilateral: A and B are vertices of the triangle ABC, so neither lies in the triangle of the other three; and if, say, D were inside ABE, the ray ED would cross the segment AB, putting A and B on opposite sides of the line DE.

Problem 3

Hint 1. Since $f \not\equiv 0$, there is x_0 with, say, $f(x_0) > 0$ (otherwise apply the argument to $-f$). Use $f \to 0$ to choose R with $|f(x)| < f(x_0)$ for $|x| > R$.

Hint 2. On the compact interval $[-R, R]$ (which contains x_0) f attains a maximum $M \geqslant f(x_0)$, and outside it $f < f(x_0) \leqslant M$; so M is the global maximum.

Hint 3. Example: $f(x) = e^{-x^2}$ attains its maximum 1, while $\inf f = 0$ is not attained.

Problem 4

Hint 1. Put the bounded set A in a cube and bisect it into 2^n subcubes: one of them contains infinitely many points of A. Iterate to get nested cubes with side $\to 0$, each containing infinitely many points of A; their common point is an accumulation point.

Hint 2. For nested non-empty compact sets $K_1 \supset K_2 \supset \dots$ choose $x_k \in K_k$. Either some value repeats infinitely often, or the set of values is infinite and bounded and has an accumulation point x. In both cases $x \in K_m$ for each m, because $x_k \in K_m$ for $k \geqslant m$ and K_m is closed.

Hint 3. Without boundedness this fails: $F_k = [k, \infty) \subset \mathbb{R}$ are nested, closed, non-empty, with empty intersection.

Problem 5

Hint 1. A finite metric space has finitely many pairs, so a pair at maximal distance exists. For non-uniqueness take the four vertices of a square (both diagonals) or any equilateral triangle.

Hint 2. For a compact metric space X, the distance $d\colon X \times X \to \mathbb{R}$ is continuous (by the triangle inequality, $|d(x,y) - d(x',y')| \leqslant d(x,x') + d(y,y')$) on the compact space $X \times X$, so it attains its supremum, the diameter.

Problem 6

Hint 1. Existence for any norm: P_n is finite-dimensional, and only polynomials with $\|p\| \leqslant 2\|f\|$ can compete with $p = 0$. This set is closed and bounded in the finite-dimensional space P_n, hence compact, and $p \mapsto \|f - p\|$ is continuous on it.

Hint 2. L^2: the norm comes from an inner product, and the best approximation is the orthogonal projection of f onto P_n. Uniqueness follows from strict convexity: if $p \neq q$ are both best, the parallelogram law shows $\|f - \frac{p+q}{2}\| < \|f - p\|$.

Hint 3. Uniform norm: it is not strictly convex, so the argument of Hint 2 fails in general (for approximation by other finite-dimensional families uniqueness may indeed fail). For polynomials uniqueness still holds, but by a special reason: if p, q are best with error E, then so is $r = \frac{p+q}{2}$, and by Chebyshev's alternation theorem $f - r$ attains $\pm E$ with alternating signs at $n + 2$ points.

Hint 4. At each of these points $f - p$ and $f - q$ must both equal the same value $\pm E$, so $p - q$ has $n + 2$ zeros and $p = q$. So: the L^2 uniqueness is a general Hilbert-space fact, the uniform one uses the Haar property of polynomials.

Problem 7

Hint 1. Since $|p(z)| \geqslant |a_n||z|^n(1 - \sum_{k<n} |a_k|/(|a_n||z|^{n-k})) \to \infty$, the infimum of $|p|$ is the same as its infimum over a large closed disc, where it is attained by compactness.

Hint 2. Let $|p|$ be minimal at z_0 and suppose $p(z_0) \neq 0$. Expand $p(z_0 + w) = p(z_0) + c_k w^k + c_{k+1} w^{k+1} + \dots$ with $c_k \neq 0$ the first non-zero coefficient after the constant one.

Hint 3. Choose ω with $c_k \omega^k = -p(z_0)$ (a k-th root exists in $\mathbb{C}$). Then $p(z_0 + t\omega) = p(z_0)(1 - t^k) + O(t^{k+1})$, so $|p(z_0 + t\omega)| < |p(z_0)|$ for small $t > 0$: a contradiction. Hence the minimum is 0.

Problem 8

Note. As stated (closed intervals) the claim is false: $[-1, 0]$ together with $[\frac{1}{k+1}, \frac{1}{k}]$, $k \geqslant 1$, cover $[0, 1]$, but no finite subfamily does. The hints follow the intended version with open intervals (open relative to $[0, 1]$).

Hint 1. Let $s = \sup\{x \in [0,1] : [0,x] \text{ is covered by finitely many members of } \mathcal{A}\}$; the set is non-empty because 0 lies in some interval.

Hint 2. Take $I \in \mathcal{A}$ containing s; it contains $(s - \varepsilon, s + \varepsilon) \cap [0, 1]$. Pick $x \in (s - \varepsilon, s]$ with $[0, x]$ finitely covered and add I: then $[0, \min(s + \varepsilon/2, 1)]$ is finitely covered. Hence $s = 1$ and the supremum is attained.

Hint 3. The standard proof argues by contradiction with bisection: if $[0, 1]$ had no finite subcover, one of the halves would not either, and so on; the nested intervals shrink to a point covered by one open interval, which then covers a whole small interval of the sequence. The sup argument is the same compactness (completeness of $\mathbb{R}$) used extremally.

Problem 9

Hint 1. Work with convex compact sets C of perimeter at most L, translated so that they contain the origin; they lie in the disc of radius $L/2$. By Blaschke's selection theorem this family is compact in the Hausdorff metric, and area and perimeter are continuous on convex sets, so a set C^* of maximal area exists. It has perimeter exactly L (otherwise enlarge it by a homothety).

Hint 2. Steiner symmetrisation S_u with respect to a line perpendicular to a direction u preserves convexity and area and does not increase the perimeter; the perimeter stays the same only if the midpoints of the chords of C parallel to u lie on one line perpendicular to u, i.e. C is already symmetric about such a line.

Hint 3. If $S_u C^*$ had smaller perimeter, a homothetic copy of it with perimeter L would have larger area than C^*. So for every direction u the set C^* has an axis of symmetry perpendicular to u. Each axis passes through the centroid, so C^* is symmetric about every line through its centroid, hence invariant under all rotations about it: a disc.

Hint 4. Existence is used in Hint 3: the argument "a non-disc can be improved" proves nothing unless a maximiser is known to exist (compare Perron's paradox: "the largest natural number is 1, since $n^2 > n$ for $n > 1$"). Steiner's original proof missed exactly this step.

Problem 10

Hint 1. (A tree here is connected.) Fix a root r. Since r has finitely many neighbours and T is infinite, one of the subtrees hanging from r is infinite.

Hint 2. Move to the root v_1 of that infinite subtree and repeat: each vertex of an infinite subtree has finitely many children, so one child carries an infinite subtree. The resulting $r, v_1, v_2, \dots$ is an infinite path.

Hint 3. Compactness, countable case: let $p_1, p_2, \dots$ be the variables and $\varphi_1, \varphi_2, \dots$ the formulas, every finite subset satisfiable. Let the vertices at level n be the truth assignments s to $p_1, \dots, p_n$ that make true every φ_i with $i \leqslant n$ whose variables are among $p_1, \dots, p_n$; the parent of s is its restriction to $p_1, \dots, p_{n-1}$.

Hint 4. Check that restriction keeps the defining property (so this is a tree), that each level is non-empty (restrict a valuation satisfying $\varphi_1, \dots, \varphi_n$), and that degrees are at most 3. An infinite path is a valuation v of all variables; each φ_i is true under v, since for $n \geqslant i$ large enough all variables of φ_i are among $p_1, \dots, p_n$.